Predictive Value of Financial-News Sentiment for Short-Term Semiconductor Sector Returns:
A Natural Language Processing Approach

Columbia University
Quantitative Methods in the Social Sciences (QMSS)
December 2025

Abstract

This study investigates whether aggregated financial-news sentiment contains short-horizon predictive information for semiconductor sector returns. Using a ten-year corpus of articles retrieved from the New York Times Article Search API and daily adjusted prices for the VanEck Semiconductor ETF (SMH), the analysis constructs a time-aligned dataset in which sentiment precedes 3-day forward market outcomes. Sentiment scores are derived from article-level tone measures using a standardized preprocessing and lexicon-based scoring pipeline, and are aggregated into daily sector sentiment indices. Predictive performance is evaluated through both regression models, which estimate forward return magnitude, and classification models, which predict directional return labels. Across all specifications, sentiment features exhibit weak or null predictive association with SMH forward returns: classifiers perform at chance levels, and regressions collapse toward mean-return predictions with negligible explanatory power. These results align with the semi-strong form of the Efficient Market Hypothesis, suggesting that general-interest news sentiment is incorporated too slowly relative to market reaction speed to yield tradable short-term signals. The findings imply that low-latency news sources or higher-frequency sentiment streams may be more appropriate for forecasting short-run movements in technology-driven sectors.

Introduction

Financial markets, especially technology-driven sectors, respond rapidly to information embedded in news narratives. This premise is grounded in Fama's Efficient Market Hypothesis, which argues that asset prices incorporate all publicly available information with high speed and efficiency (Fama, 1970). Within this framework, price movements in efficient markets should rapidly reflect publicly available information. Consequently, signals that are overly simple or fully public face strong competitive trading pressure, because market participants can observe, extract, and act on them almost instantaneously. As a result, markets tend to eliminate the return advantage of these signals through rapid arbitrage, a process accelerated when many actors react to the same information at nearly the same time (Fama, 1970, pp. 383-386).

Nevertheless, a growing body of empirical research suggests that richer sentiment measures derived from textual analysis of news media may still exert a measurable influence on short-term stock returns. For example, negative media tone extracted from financial news shows significant association with next-day to multi-day excess returns, functioning as a priced risk or attention factor rather than random noise (Bask et al., 2024, p. 303). Another investigation indicates that news sentiment affects returns on a short-term horizon: daily or multi-day sentiment changes predict subsequent stock performance, while weekly aggregated sentiment can forecast return patterns over longer horizons (Heston and Sinham, 2016). These findings motivate the central question of this study: **does aggregate financial-news sentiment predict short-term returns in the semiconductor technology sector?** By focusing on a sector

characterized by high sensitivity to news flow, rapid information diffusion, and frequent supply-chain as well as regulatory developments, this research aims to test whether advanced sentiment metrics capture predictive content beyond what markets immediately price in.

Literature Review

Building on the Efficient Market Hypothesis (EMH) benchmark introduced above, this section reviews prior work on (i) investor sentiment and short-run return predictability, (ii) text-based sentiment measures from news and social media, and (iii) evidence for amplified sentiment effects in technology-intensive sectors. Together, these strands motivate our focus on aggregated financial news sentiment and short-horizon semiconductor ETF returns.

Classical EMH implies that, in frictionless markets with rational investors, publicly available information should be incorporated into prices almost instantaneously (Fama, 1970). Behavioral finance relaxes these assumptions and shows that investor beliefs, emotions, and limits to arbitrage can generate temporary deviations from fundamental values, especially over short horizons (Shiller, 2003; Barberis, Shleifer, & Vishny, 1998). Early empirical studies construct survey- or market-based sentiment indices and document that these indices covary with and sometimes forecast near-term returns. Brown and Cliff (2004) find that aggregate sentiment is related to contemporaneous and subsequent market performance, while Baker and Wurgler (2006) show that sentiment particularly affects hard-to-value, speculative stocks. These results indicate that sentiment is a non-fundamental state variable that can produce short-lived predictability, creating a tension with the strong-form EMH benchmark that our study also exploits.

A second wave of research moves from coarse sentiment indices to text-based measures that extract tone directly from written narratives. Tetlock (2007) asserts that pessimistic language in the daily content from a popular *Wall Street Journal* column predicts lower next-day market returns, and an unusual level of pessimism predicts high market trading volume, consistent with temporary overreaction. Subsequent studies incorporate richer news sources and more sophisticated text-analysis tools, finding that news tone, topic structure, and information volume help explain returns, volatility, and trading activity over horizons of one to several days. In this literature, sentiment is often interpreted as an attention or information-processing factor: negative or surprising news attracts attention, shifts risk perceptions, and is suspected to generate short-run return patterns before being arbitraged away.

Parallel work uses social media sentiment (e.g., Twitter, StockTwits, Reddit) as an alternative high-frequency signal of investor mood. Large-scale studies show that the emotional tone and volume of posts are associated with market movements, trading intensity, and volatility (Bollen, Mao, & Zeng, 2011; Sprenger et al., 2014). More recent applications combine social-media sentiment with machine-learning models: Kolasani and Assaf (2020), for example, first train a sentiment classifier on Twitter data and then use the resulting daily sentiment scores

to predict Apple (AAPL) and Dow Jones Industrial Average price changes with boosted regression trees and neural networks, finding that the neural network model outperforms simpler benchmarks. Meanwhile, Zhang, et al. (2011) assert that social media sentiment is often more informative for volatility or order flow than for directional returns, and that its predictive content can be noisy and unstable.

The technology sector provides a natural laboratory for examining these effects. Tech and semiconductor firms are highly sensitive to information about innovation, regulation, supply-chain disruptions, and geopolitical risk, and their valuations often embed substantial growth and option-like components.

Hypothesis

Grounded in the Fama's Efficient Market Hypothesis theory, which asserts that prices rapidly incorporate public information, this study tests whether aggregated financial-news sentiment contains short-term predictive content that markets do not immediately arbitrage away due to competitive and near-simultaneous trading responses. Prior work shows that rolling news sentiment predicts 1 to 5 day forward returns in information-sensitive technology industries, and that predictive power decays when using static or longer aggregation windows (Heston and Sinham, 2016, pp. 220–224). Therefore, this study proposes the central research hypothesis: Aggregate financial-news sentiment is a significant predictor of 1 to 3 day forward semiconductor sector returns.

Data

This study relies on two complementary data streams: (1) a financial-news text corpus sourced from the New York Times Article Search API, and (2) semiconductor-sector market returns represented by the VanEck Semiconductor ETF (SMH), extracted as daily adjusted price data from Yahoo Finance. Together, these sources allow the project to evaluate whether narrative sentiment encoded in financial media correlates with short-horizon sector index movements.

The first data stream is collected from the New York Times Article Search API, a structured archive that provides both article text and descriptive metadata, including the title, abstract, snippet, publication date, section tags, and keywords. The query is conducted using the keywords Semiconductor, Tech, and Stocks, producing a 10-year media window spanning September 2015 to September 2025. Because the API delivers labeled sections and summaries along with article text, it enables the construction of a narrative-sensitive dataset for sector-level sentiment extraction, rather than headline-attention or surface-frequency proxies alone.

To ensure sector relevance, articles were filtered based on high information density in financial and technological discourse. The Business section contributes 1,057 articles, followed by Technology (917), U.S. (392), and World (199), together representing the largest,

market-relevant slices of the full retrieval set. Additional sections such as Briefing, Opinion, and Science are retained to preserve market-adjacent narrative framing, including policy discourse, supply-chain disruptions, consumer-technology dependencies, and global semiconductor ecosystem stressors that may influence investor expectations at the sector index level.

The second data stream consists of daily, end-of-day adjusted returns for the SMH ETF, derived from its adjusted price series on Yahoo Finance. To remain aligned with short-horizon analysis, sector return direction for 1 to 5 day forward windows is computed exclusively from adjusted closing prices, ensuring that observed market reactions reflect sector-level movement rather than raw price distortions due to splits or dividends.

The VanEck Semiconductor ETF (SMH) targets semiconductor-sector performance by tracking, as operationally closely as feasible before fund-level fees and expenses, the MVIS US Listed Semiconductor 25 Index (MVSMHTR). This index measures the aggregated performance of United States listed firms engaged in semiconductor manufacturing, integrated-circuit production, and the equipment, infrastructure, and industrial supply systems that support chip fabrication and distribution (VanEck). Because SMH offers high liquidity, institutional and retail tradability, transparent pricing, and heightened sensitivity to narrative shocks, it is used as the primary market-response variable for testing sector-level sentiment-to-return alignment over short forecast horizons.

Methodology

This research operationalizes sector-level narrative effects by constructing a time-aligned semiconductor news corpus and evaluating its tone against short-horizon forward returns of the VanEck Semiconductor ETF (SMH). The workflow is sequenced chronologically to ensure that sentiment inference strictly precedes return-windown evaluation, maintaining temporal validity under the Efficient Market Hypothesis framework (Fama, 1970). Within this framework, the modeling objective is twofold: (1) to generate a structured NLP-derived sentiment signal from financial news and (2) to test whether that signal exhibits short-horizon directional linkage with semiconductor-sector price movement, operationalized as directional return prediction and forward correlation significance testing across 1 to 5 trading days, with an emphasis on 3-day return direction.

Data and Narrative Filtering

Raw financial media data collected through scripted HTTP requests to the New York Times Article Search API using Python's requests library. Articles are retrieved iteratively in paginated form, parsed from the JSON response payload, and converted into a pandas DataFrame. To ensure offline processing and lag-based return matching, retrieved articles are stored locally as a CSV file rather than streamed dynamically during modeling tasks. This

permanent local storage enables sentiment modeling without real-time API calls contaminating train/test partitions.

The forward market-response target is represented by the SMH ETF, whose adjusted closing prices are extracted using Yahoo Finance price endpoints and aligned exclusively at an end-of-day trading resolution. Using adjusted closing prices ensures that return labels reflect normalized economic price change rather than distortions from splits or dividend adjustments.

Text Preprocessing

The textual signal is engineered through a multi-stage article-level preprocessing pipeline designed to balance linguistic fidelity with temporal and statistical validation for short-horizon financial inference. Each news article is first decomposed via tokenization, converting unstructured media narratives into discrete lexical units while preserving term boundaries necessary for sequence-independent tone scoring. All tokens are converted to lowercase to enforce case invariance, a normalization step that reduces feature sparsity and prevents artificial vocabulary duplication. High-frequency but semantically uninformative terms are filtered through stop-word removal, mitigating noise amplification from non-market language that could bias sentiment aggregates toward general policy discourse rather than economically relevant tone. To preserve semantic consistency beyond surface-form cleaning, a context-aware transformation layer, implemented through utils.xform_fun, maps morphological variants into shared semantic representations, reducing dispersion in finance narratives without injecting future information. Sentiment scoring is then conducted on the fully lowercased article text using SentimentIntensityAnalyzer() from VADER, ensuring that tone extraction operates on a standardized lexical space and is not distorted by case-based sparsity or duplicate token dimensions. Additionally, section tags and article summary metadata are preserved in the cleaned corpus to maintain document-level narrative scaffolding, ensuring that sentiment aggregation reflects structured media information rather than collapsed text streams, supporting both hierarchical corpus inference and rolling tone-dynamic capture used for correlation testing.

Corpus Indexing and Temporal Alignment

The sentiment corpus is indexed by article publication timestamp t, creating an ordered narrative ledger that contains the casual sequencing required for evaluating public information incorporation in financial prices. On each trading day t, only articles published prior to market close are chosen into the feature pipeline, ensuring that the model does not encode narratives formed after prices are realized. To estimate tone dynamics without look-ahead distractions, a rolling interval $[t-T, t-1]$ is enforced, where T denotes that the retrospective depth measured in trading days; this allows the model to condition learning on lag-bounded semantic context rather than contemporaneous price co-movement, improving inferential purity when testing short-horizon sentiment-price relationships. Market response is represented using SMH adjusted closing prices, which are shifted forward relative to the corpus index so that sentiment

observations always precede price movement windows. Directional labels are then constructed to reflect pure next-day sector movement, defined as:

$$Y_t = 1 \; if \; closing \; price_t > closing \; price_{t-1}, \; Y_t = 0 \; otherwise.$$

This label schema ensures that predictions and correlation tests evaluate forward market directionality rather than instantaneous or within-day reactions. By imposing this forward indexing constraint, the study eliminates temporal leakage and guarantees that any measured sentiment-return linkage reflects predictive media-information assimilation into next-day prices, rather than labels generated in reaction to price movement itself.

Sentiment Modeling and Sentiment Signal Aggregation

Sentiment is derived using VADER's SentimentIntensityAnalyzer, a lexicon- and rule-based model optimized for short, noisy financial media narratives, which produces continuous numeric tone estimates rather than learned probabilistic labels. The pipeline extracts the four VADER polarity components (pos, neu, neg) for internal tone decomposition, while the compound score, bounded in [-1, 1], is adopted as the primary signed sentiment intensity metric, where positive values encode optimistic or upward-leaning sector narratives, negative values encode pessimistic or downward-leaning narratives, and values near zero reflect neutral or mixed tone, preserving the gradation of tone strength necessary for return-direction inference (Hutto and Gilbert, 2014, pp. 216-225). These article-level sentiment scores are time-stamped at publication day *t* and aggregated only from narratives appearing strictly before daily market close, maintaining causal validity for prediction windows. Daily sector sentiment is computed as the rolling mean of article-level compound scores, allowing final downstream signal (sent_3d or similar series) to encode rolling tone dynamics rather than static proxies, producing a continuous daily semiconductor-sector sentiment index series that is merged forward into ETF return modeling tasks as a predictive feature rather than a neural training target, ensuring that return classifiers exclusively predict future SMH directional movement (y_3d) conditioned on past sentiment, not sentiment itself.

Dual Modeling Rationale and Model Validation

To comprehensively evaluate the sentiment-return relationship, the study adopts a dual modeling strategy, combining regression and classification learners to test for both continuous return magnitude and directional co-movement. Regression models estimate sentiment-return linkage on continuous 3-day forward return values (ret_3d), quantifying whether tone intensity explains return magnitude and mispricing amplitude, while classification models condition on binary directional labels (y_3d), measuring whether sentiment carries a separable signal for return polarity rather than scale alone. This approach supports hypothesis testing under two complementary financial viewpoints: the underreaction/gradual-pricing perspective, where narratives may explain magnitude before full incorporation, and the information-efficiency perspective, in which direction is expected to incorporate faster than size. By training classifiers

on future return direction rather than sentiment, the study ensures that the model capacity is used only to learn sentiment-conditioned ETF directionality, not sentiment itself. Model Validation is conducted on a strict temporal train/test split, preventing both look-ahead and label leakage. Predictive performance is quantified using Receiver Operating Characteristic – Area Under the Curve (ROC–AUC), weighted precision, recall, and the F1 score to evaluate the return-direction classification model, and Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-square for the continuous return regressors, following established standards for short-horizon financial media prediction (Hutto & Gilbert, 2014, pp. 216-225). This validation architecture enables inference on whether sentiment exhibits economic relevance through direction, magnitude, or both, while explicitly controlling for temporal leakage and short-horizon market microstructure noise.

Results

The empirical analysis evaluated whether aggregated financial-news sentiment provided predictive information about 3-day forward returns in the semiconductor sector. Across both the classification and regression frameworks, the evidence consistently indicates that sentiment derived from New York Times coverage does not contain usable short-horizon return information. Performance levels across all modeling strategies remained statistically indistinguishable from random chance, suggesting an absence of systematic sentiment–return linkage.

Classification Performance

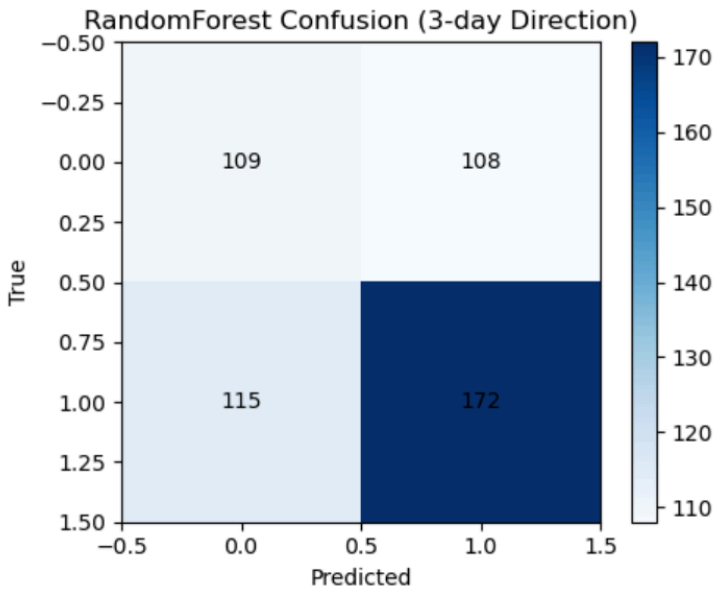
Figure 1: Summary Table of Classification Metrics

Model	**Accuracy**	**Precision**	**Recall**	**F1 Score**
Logistic (balanced)	0.494	0.511	0.494	0.495
Random Forest	0.558	0.559	0.558	0.558
Gradient Boosting	0.548	0.518	0.548	0.503

In the classification task (predicting upward vs. downward movement), three models were employed: Penalized Logistic Regression, Random Forest Classifier, and Gradient Boosting Classifier. Performance metrics across all three models revealed a lack of signal separability. The precision and recall scores exhibited only marginal variance between models, clustering near 0.50, which suggests performance equivalent to random chance. Specifically, the

F1 weighted scores were 0.49 for Penalized Logistic Regression, 0.50 for Gradient Boosting, and 0.56 for the Random Forest Classifier. While the Random Forest model achieved a slight improvement over the baseline, the lift is likely attributable to overfitting or noise rather than a robust capture of market dynamics.

Figure 2: Confusion Matrices for All Classifiers



The confusion matrix analysis further strengthens the conclusion that the sentiment features did not encode information relevant for short-horizon return direction. Although the Random Forest classifier generated a nontrivial distribution of predictions, anticipating downward movement in 224 cases and upward movement in 280 cases, this variation reflects model mechanics rather than meaningful pattern recognition. The predictive allocations did not correspond to actual market outcomes in a statistically informative way. For instance, on days when the true market movement was negative, the model classified 109 observations correctly but misclassified 108, yielding an essentially even split between true negatives and false positives. This symmetry indicates that the classifier failed to identify any stable feature patterns that differentiate negative-return days from positive-return days.

The behavior of the model therefore resembles stochastic guessing rather than systematic inference: the model's allocation of predictions across labels appears unrelated to underlying return realizations. The near-equal rates of correct and incorrect predictions within each actual class suggest that the classifier is responding predominantly to noise in the sentiment data rather than to economically meaningful variation. This outcome is consistent with scenarios in which the predictors lack directional signal, causing the decision boundaries learned by the model to fluctuate around random partitions of the data rather than converging toward a discriminative structure.

Figure 3: ROC Curves

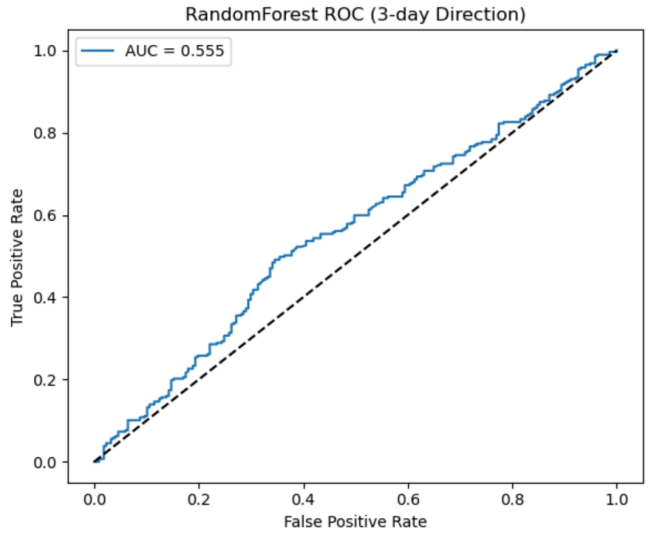
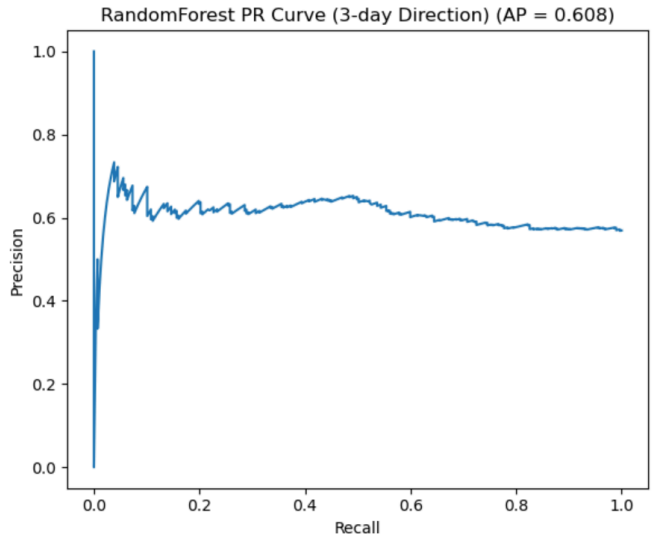


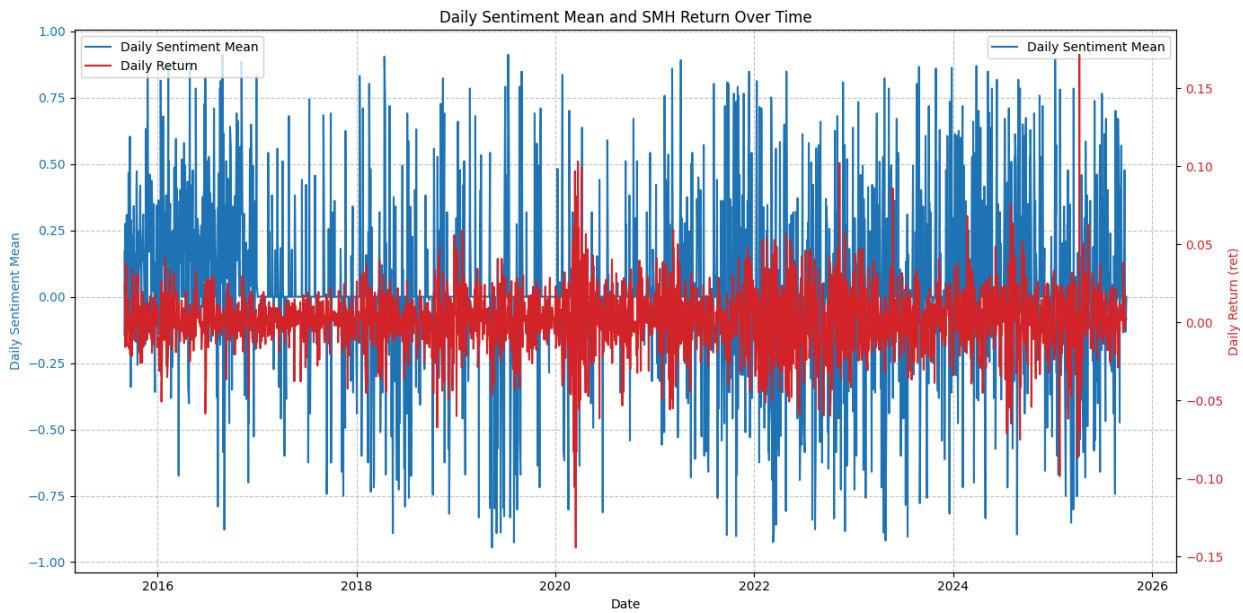
Figure 4: Precision–Recall Curves



The ROC and Precision–Recall diagnostics provide further evidence that the sentiment features do not contain discriminative information for short-horizon return direction. The Receiver Operating Characteristic (ROC) curves for all classifiers remain close to the 45-degree diagonal, with AUC values between 0.50 and 0.56. An AUC in this range indicates that, across all possible classification thresholds, the models do not rank positive-return observations higher than negative-return observations in a consistent or statistically meaningful way. In practical terms, the classifiers are unable to assign higher predicted probabilities to true positive cases at rates exceeding random ordering.

The Precision–Recall (PR) curves exhibit a similarly flat profile. Precision does not increase noticeably as recall improves, and the curves show no concentration of points in regions associated with strong signal detection. This pattern reflects the underlying imbalance between model confidence and correctness: when the classifiers identify instances as high-probability positives, the precision achieved in those regions does not exceed baseline levels. Hence, the PR curves corroborate the ROC findings by demonstrating that the models cannot isolate subsets of predictions with reliably higher correctness.

Figure 5: Time-Series Plot of Daily Sentiment vs. Forward SMH Returns



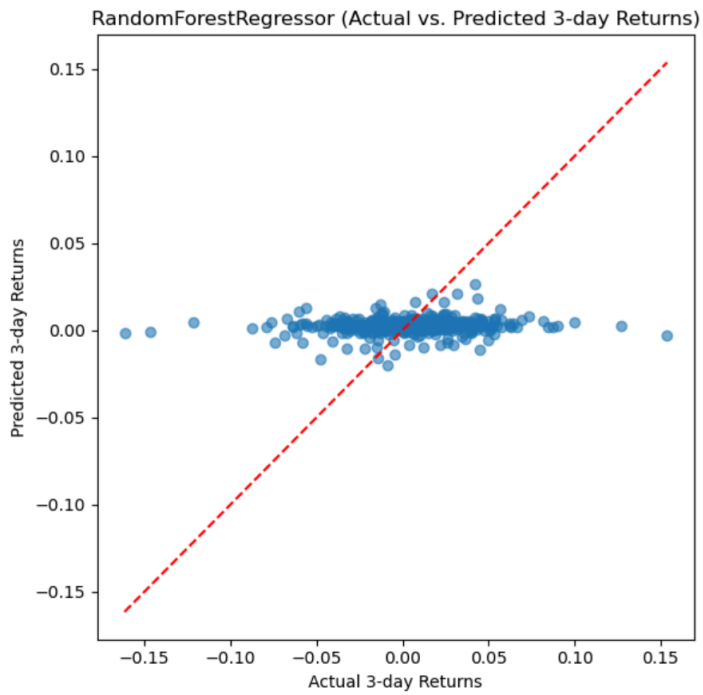
The time-series plot of Daily Sentiment Mean and SMH Return Over Time shows that the sentiment index fluctuates at a much larger amplitude than the 3-day forward SMH return series, while the return line remains centered tightly around zero with fewer excursions. This visual divergence indicates that sentiment intensity is high-variance even on days when the semiconductor sector exhibits minimal 3-day directional movement, implying poor temporal coherence between media tone and future price shifts. The lack of synchronized sentiment–return inflection points provides a logical explanation for the earlier classification results being near-random, and for regressors failing to track realized return dispersion.

Regression Performance

The regression analysis, which attempted to predict the exact magnitude of 3-day returns (ret_3d), yielded Mean Squared Error (MSE) values of 0.0011 for Gradient Boosting and 0.0010

for both Random Forest and Ridge Regression. While these MSE values appear superficially low, they are deceptive due to the scale of the target variable (decimal percentage returns). When converted to the Root Mean Squared Error (RMSE), the average error is approximately 0.03 (3.0%). Given that the average absolute 3-day return in the test set is approximately 0.0031 (0.31%), the prediction error exceeds the average signal magnitude by a factor of ten.

Figure 6: Actual vs. Predicted Return Scatterplot



The visualization of Actual vs. Predicted Returns illustrates a phenomenon known as "collapsing to the mean." The predicted values form a horizontal cluster near zero, failing to capture the variance of the actual returns, which range from -15% to +15%. This geometric pattern indicates that the models could not find a linear or non-linear relationship between NYT sentiment and price magnitude. To minimize the loss function in the absence of a strong signal, the models converged on predicting the historical mean return (effectively zero) for every observation, resulting in a low MSE but zero practical predictive capability.

Conclusion

This study aimed to determine if narrative sentiment extracted from the New York Times could serve as a leading indicator for short-term semiconductor sector returns. Based on the empirical results across both classification and regression tasks, we fail to reject the null hypothesis. The data suggests that aggregate financial news sentiment, as proxied by general-interest media, does not possess statistically significant predictive power for 1 to 3-day price movements in the semiconductor industry.

These findings strongly align with the semi-strong form of Fama's Efficient Market Hypothesis (1970), which posits that asset prices adjust rapidly to all publicly available information. The lack of predictive signal likely stems from the latency of the data source relative to market action. By the time a narrative is synthesized, edited, and published in a major outlet like the New York Times, the information content has likely already been arbitraged by high-frequency algorithmic traders reacting to real-time newswires (e.g., Bloomberg or Reuters) milliseconds after the initial event.

Therefore, while the New York Times effectively captures the state of public discourse and may reflect broader economic anxiety or optimism, it functions as a lagging indicator of market sentiment rather than a leading predictor of asset pricing. Future research might achieve better predictive success by utilizing lower-latency data sources, such as raw newswires or social media feeds, or by focusing on alternative target variables such as volatility or trading volume, which may be more sensitive to mass-media attention than price direction itself.

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